# Your Results, From Zero

For Student 2 · everything explained step by step, with our own numbers

You do not need to be good at statistics to present the Results. You need to understand **one idea per table or figure** and be able to say it simply. This guide teaches you exactly that.

## How to use this guide (3 sittings)

1. **Sitting 1 (1.5 h):** Part A (the story of the results) and Part B (statistics lessons). Read slowly. Say every "Say" box out loud.
2. **Sitting 2 (2 h):** Part C (why each table and figure exists, how to read it). Have the thesis open next to you.
3. **Sitting 3 (1.5 h):** Part D (your slides: what to point at), Part E (drills). Then do the drills again the next day without looking.

Keep the one-page cheat sheet (last page) in your pocket for the week before the defence.

## Part A · The Results tell one story in nine steps

Every table and figure in Chapter 3 answers one question. They come in a logical order: first **who** answered, then **what they do**, then **how anxious** they are, then the **main link**, then **what else** could explain it, then **who is at risk**, then **behaviour**, then **pathways**, and finally **is it robust?**

| Step | Question | Short answer | Where | Slide |
|---|---|---|---|---|
| 1 | Who took part? | 300 young, educated adults; 44.0% medical students | Table 1 | 14 |
| 2 | How do they search? | AI chatbots first (78.0%), search engines second (56.3%) | Table 2, Figure 3 | 15 |
| 3 | What do they do after searching? | 34.7% delay care when reassured; 31.7% never tell the doctor | Figure 4 | 22 |
| 4 | How anxious are they? Do the scales work? | 19% elevated; both scales reliable; CSS-12 same in Arabic and English | Tables 3, 4, 5 | 16 |
| 5 | Are health anxiety and cyberchondria linked? (H1) | Yes, r = 0.47 | Table 6, Figures 5, 6, 11 | 17, 19 |
| 6 | Which groups score higher? | Chronic illness, reassurance seekers; AI users (but see step 7) | Tables 7, 8, Figure 7 | – |
| 7 | Does the link survive other factors? (H2) Who is at risk? | Yes; health anxiety adds 18.4%; AI effect not robust | Tables 9, 10, Figure 8 | 18, 20, 23 |
| 8 | Does it affect behaviour? (H4) | Cyberchondria → more disclosure (robust); others exploratory | Table 11 | 22 |
| 9 | Is there a cycle? (H3) Is everything robust? | Compatible with a cycle, not proven; all checks unchanged | Table 12, Figures 9, 10, Table 13 | 21 |

> **Say:** *Our results answer nine questions in order. In short: AI chatbots are the main source of health information, one in five has elevated health anxiety, and health anxiety and cyberchondria are clearly linked, r = 0.47, even after adjusting for everything else.*

# Part B · Statistics from zero (with our numbers)

Each lesson: the idea in plain words, our example, and what to say. Read them in order; each one builds on the one before.

### 1. Variables: two kinds

- **Categorical** = groups (gender, uses AI yes/no, tool profile). We describe them with **n and %**.
- **Continuous** = scores on a scale (SHAI 0–42, CSS 12–60). We describe them with **mean, SD, median**.

This one fact decides almost every choice of test and chart in the thesis.

### 2. Mean and SD

**Mean** = the average. **SD (standard deviation)** = how spread out people are around the average. About two-thirds of people lie within 1 SD of the mean.

```
SHAI mean 12.3 ± SD 5.8 → most people scored between about 6.5 and 18
```

> **Say:** *The average health-anxiety score was 12.3 out of 42.*

### 3. Median and IQR

The **median** is the middle person. The **IQR** is the range of the middle half. We use them when scores are skewed (not a symmetrical bell shape).

```
SHAI median 12 (IQR 8–16): half of people scored 8 to 16
```

### 4. Normal distribution and why it matters

Many tests assume scores form a bell curve. The **Shapiro–Wilk test** checks this. Our scores were slightly skewed ($p < 0.01$), so we used **non-parametric** tests (they do not need the bell shape) and **robust** regression.

> **Say:** *Our scores were not perfectly normal, so we used tests that do not assume normality.*

### 5. p-value

Imagine there is truly **no** link. The p-value is the chance of seeing a result as strong as ours just by luck. Small p (below 0.05) = "unlikely to be luck" = **significant**. It does **not** say how big or important the effect is.

```
r = 0.47, p < 0.001 → almost impossible to get this by chance if there were no
link
```

> **Trap:** p tells you if an effect exists, not how big it is. For size, look at r, β, OR.

### 6. 95% confidence interval (CI)

Our number is an estimate from 300 people. The CI is the range where the true value probably lies. Narrow CI = precise. If a CI for an odds ratio includes 1, the result is not significant.

```
r = 0.47, 95% CI 0.38–0.55 → the true correlation is probably between 0.38 and 0.55: always clearly above zero
```

### 7. Correlation r

How closely two scores move together: 0 = no link, 0.1 small, 0.3 moderate, 0.5 large, 1 = perfect. Positive = both go up together.

```
r = 0.47: people with more health anxiety tend to have more cyberchondria. r² = 22%: health anxiety explains about a fifth of the differences in cyberchondria.
```

> **Say:** *Health anxiety and cyberchondria rise together: r = 0.47, a moderate link.*

### 8. Pearson vs Spearman

**Pearson r** assumes a straight-line link; **Spearman ρ** uses ranks and does not need normal data. We report both; both were 0.47, so the result does not depend on the method.

### 9. Partial correlation

The same correlation after "removing" other variables (age, gender, background, chronic disease, trust, AI use). If it stays the same, those variables are not the explanation.

```
r = 0.47 → partial r = 0.48: unchanged
```

> **Say:** *The link stayed the same after adjusting for everything else, so it is not explained by age, gender or background.*

### 10. Comparing groups: Mann–Whitney and Kruskal–Wallis

To compare a **score** between groups without assuming normality: **Mann–Whitney U** for 2 groups, **Kruskal–Wallis H** for 3 or more.

```
Chronic condition: SHAI 14.2 vs 11.9 → Mann–Whitney p = 0.004
```

### 11. Chi-square ($\chi^2$)

To test if two **categorical** variables are related, e.g. "elevated health anxiety yes/no" and "uses social media yes/no". It compares the counts we saw with the counts we would expect if there were no link.

### 12. Many tests → FDR correction (q-value)

If you run 162 tests at p < 0.05, about 8 will be "significant" by pure luck (162 × 0.05 ≈ 8). The **Benjamini–Hochberg FDR** correction adjusts p-values into **q-values**. q < 0.05 = **robust**; p < 0.05 but q ≥ 0.05 = **exploratory** (could be luck).

```
Never disclosing: CSS q = 0.012 (robust) · SHAI q = 0.098 (exploratory)
```

> **Say:** *Because we ran many tests, we corrected for multiple testing; results that did not survive are called exploratory.*

### 13. Linear regression, B and β

Regression predicts a score (CSS) from several factors at the same time, so each factor is "adjusted" for the others. **B** = how many CSS points change per 1 unit of the factor. **β** = the same on a common scale, so you can compare factors (bigger β = stronger).

```
SHAI: B = 0.68, β = 0.48. Someone with SHAI 20 vs 10 → about 6.8 CSS points
higher.
```

> **Say:** *Each extra point of health anxiety adds about 0.7 points of cyberchondria; it is by far the strongest factor.*

### 14. $R^2$, $\Delta R^2$ and hierarchical regression

**$R^2$** = share of differences the model explains (0–1). In **hierarchical** regression we add groups of variables in steps and see how much each step adds (**$\Delta R^2$**).

```
Step 1 demographics R² 0.030 (3%) → step 2 digital use 0.133 (13%) → step 3
health anxiety 0.317 (32%). Health anxiety adds 18.4%.
```

> **Why this was chosen:** Hierarchical order shows that health anxiety matters even after everything else is already in the model.

### 15. Robust standard errors (HC3)

Regression assumes the "errors" are equally spread. Ours were not (Breusch–Pagan p = 0.004). HC3 robust standard errors fix the p-values and CIs so they are still valid. You only need: "we used robust standard errors because the residuals were uneven".

## 16. Odds and odds ratios (logistic regression)

**Logistic regression** predicts a yes/no outcome (elevated health anxiety yes/no). It gives **odds ratios (OR)**: OR 1 = no effect, above 1 = higher odds, below 1 = lower odds. **Adjusted** OR (AOR) = after accounting for the other factors.

```
Worked example from our data:
With elevated health anxiety: 52.6% had high cyberchondria → odds = 52.6/47.4 =
1.11
Without: 23.0% → odds = 23.0/77.0 = 0.30
Odds ratio = 1.11 / 0.30 ≈ 3.7
```

In the model: AOR **3.12** per **1 SD** of cyberchondria (1 SD ≈ 8.4 CSS points) means about three times the odds for every 8 extra points.

> **Trap:** AOR below 1 means protective. Age AOR 0.61: each older age group had lower odds.

## 17. AUC

How well the logistic model separates people with and without elevated health anxiety. Pick one person with and one without: AUC is the chance the model gives the higher risk to the right one. 0.5 = coin toss, 1 = perfect.

```
AUC 0.83, and 0.79 after correcting for optimism (bootstrap): good
```

> **Why this was chosen:** A model always looks better on the data it was built on, so we corrected the AUC with 500 bootstrap repeats.

## 18. Bootstrap

A computer trick: draw 5000 new samples of 300 people from our own data (with replacement), redo the analysis each time, and see how much the result varies. That gives a CI without assuming normality. We used it for mediation and to check the regression coefficient.

## 19. Mediation (indirect effect)

Does part of the link from X to Y go **through** a middle variable M? Path **a** = X → M; path **b** = M → Y (adjusted for X); **indirect** = a × b; **direct** (c′) = what is left.

```
Model A: health anxiety → excessive searching → distress
a = 0.46, b = 0.35 → indirect = 0.46 × 0.35 ≈ 0.16
total = 0.46, direct = 0.30 → about 35% goes through excessive searching
Model C: health anxiety → distress → reassurance seeking: indirect 0.22, direct
0.01 → almost all goes through distress
```

> **Trap:** Mediation in one survey is NOT proof of cause. The reverse order was also significant (0.15). Always say: compatible with a cycle, not proof.

## 20. Moderation (interaction)

Does a third variable change the **strength** of the link? E.g. is the SHAI–CSS link stronger in AI users? We tested 7 moderators; none were significant after FDR. So the link is similar in everyone.

### 21. Reliability: Cronbach α and McDonald ω

Do the items of a scale measure the same thing? 0.7 acceptable, 0.8 good, 0.9 excellent.

```
SHAI α 0.81, CSS α 0.86: good
```

### 22. CFA and fit indices

**Confirmatory factor analysis** checks whether the items group into the factors the scale claims (CSS-12 = four facets). Fit indices: **CFI** near 0.95 or higher = good; **RMSEA** and **SRMR** 0.08 or lower = acceptable.

```
CSS-12 four factors: CFI 0.95, RMSEA 0.070, SRMR 0.053 → acceptable · SHAI one
factor: CFI 0.83 → poor (it has two factors)
```

### 23. Measurement invariance

Before comparing Arabic and English answers, we must check the questionnaire works the same in both, like checking two rulers measure the same centimetre. Three steps: **configural** (same shape), **metric** (items equally strong), **scalar** (same starting points). If fit does not get worse at a step (CFI drop ≤ 0.01), that step passes.

CSS-12 passed all three. SHAI-14 passed the first two.

### 24. DIF

Differential item functioning: does a **single question** behave differently in Arabic vs English for people with the same total score? None did after FDR correction.

### 25. Sensitivity analysis

Re-doing the main analysis in another way (excluding doubtful responses, removing one item, another method) to see if the result changes. Ours did not change: r stayed 0.46–0.48.

# Part C · Every table and figure: why it is there and how to read it

## The rules that decided every choice

| If the data are... | ...we used | Example |
|---|---|---|
| Exact numbers that people need to look up | a **table** | Table 1 participants, Table 9 regression |
| A pattern people should see at a glance | a **figure** | Figure 5 scatter plot |
| Percentages of groups (categorical) | **bar chart** | Figures 3, 4 |
| Two continuous scores together | **scatter plot** | Figure 5 |
| Many correlations at once | **heatmap** (colour = strength) | Figures 6, 11 |
| A score compared across groups, skewed data | **box plot** (median, spread, outliers) | Figure 7 |
| Means of several facets in two groups | **bar chart with 95% CI** | Figure 10 |
| Many odds ratios with CIs | **forest plot** | Figure 8 |
| Paths between variables | **path diagram** | Figure 9 |

Also: the most important results get **both** a table (exact numbers) and a figure (the picture), e.g. the main correlation (Table 6 + Figure 5) and the logistic model (Table 10 + Figure 8). Less central results have only a table.

### Table 1 · Who took part

> **Why this was chosen:** Every study must describe its sample first, so the reader knows who the results apply to. A table, because the reader needs exact n and %.

**How to read it:**

1. Rows = characteristics (age, gender, education, study/work, background, chronic disease, language).
2. Columns = n (number of people) and %.

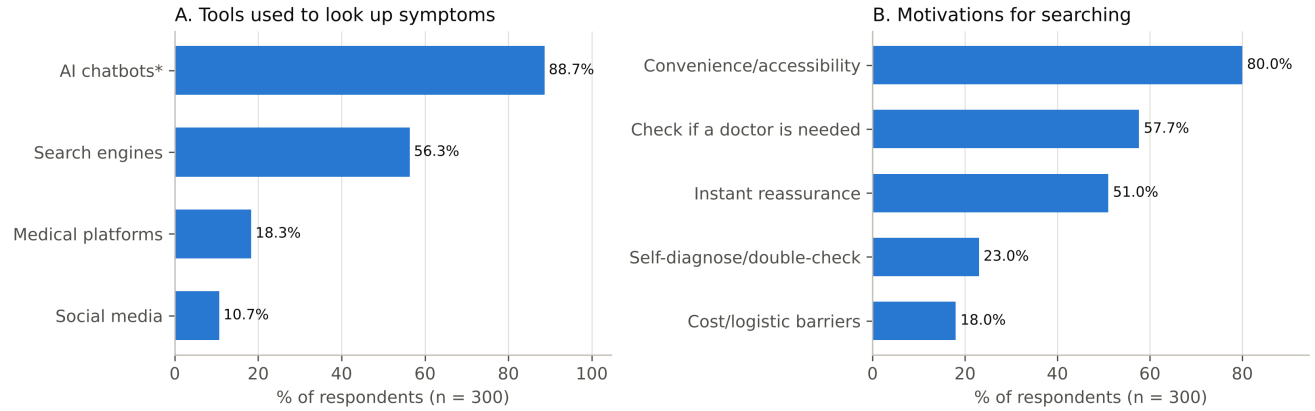
**What it shows:** A young sample (62.3% aged 20–29), 44.0% medical students, 63.3% with a health background.

> **Say:** *Our 300 participants were mostly young and educated; almost half were medical students. This matters when we interpret percentages.*

> **Trap:** Why so many medical students? Answer: the link spread through our own networks first; we say percentages cannot be generalised.

### Table 2 + Figure 3 · How people search and why

> **Why this was chosen:** Tools and motives are categories with multiple answers, so percentages are best shown as bars (Figure 3); Table 2 keeps every exact count.




**How to read it:**

1. Figure 3, panel A = tools; panel B = motives.
2. Each bar = % of all 300 who ticked that option.
3. Bars add up to more than 100% because people could tick several.

**What it shows:** AI chatbots 78.0% ticked (88.7% including those who named one) vs search engines 56.3%; convenience is the top motive.

> **Say:** *AI chatbots are now the main gateway: 78.0% ticked them, compared with 56.3% for search engines.*

> **Trap:** Why two AI numbers? 32 people named a chatbot but did not tick "AI"; we report both and test both.

## Figure 4 · What people do after searching

> **Why this was chosen:** Three related sets of categories (behaviour, disclosure, relationship) shown side by side so the committee sees the whole consultation story in one picture.




**How to read it:**

1. Panel A: behaviours after searching (multiple answers).
2. Panel B: do they tell the doctor? (always / sometimes / never).
3. Panel C: effect on the relationship with doctors.

**What it shows:** 34.7% delay care when reassured; 31.7% never tell the doctor; 34.7% say trust was strained.

> **Say:** *Searching changes the consultation: one in three delays care because they feel reassured, and almost one in three never tells the doctor.*

## Table 3 · Scores and reliability

> **Why this was chosen:** Before using any score, we must show the scales work (reliability) and give their levels. A table because these are exact values.

**How to read it:**

1. Rows: SHAI, CSS and the four CSS facets.
2. Columns: mean ± SD, median (IQR), range, α (95% CI), ω.
3. The note under the table has the CFA fit.

**What it shows:** Both scales reliable (α 0.81 and 0.86); excessiveness is the highest facet, compulsion the lowest.

> **Say:** *Both questionnaires were reliable. People search a lot, but it rarely disrupts their daily life.*

## Table 4 · Prevalence at different cut-offs

> **Why this was chosen:** The % with "elevated" health anxiety depends on the cut-off you choose. Showing four cut-offs is honest and stops the committee from accusing us of choosing the one that looks best.

**How to read it:**

1. Each row = a cut-off (15, 18, 22, 29).
2. Columns: n, %, 95% CI.
3. Bold row = 18, the one we use.

**What it shows:** 32.7% at 15, 19.0% at 18, 7.3% at 22, 0.7% at 29.

> **Say:** *About one in five scored 18 or more; this is a screening threshold, not a diagnosis, and the table shows how the estimate changes with the cut-off.*

> **Trap:** Why 18? It best separated people with health anxiety from healthy controls in the Swedish study.

## Table 5 · Arabic vs English (measurement invariance)

> **Why this was chosen:** We combined Arabic and English answers. To do that fairly, we had to prove the questionnaire works the same in both languages.

**How to read it:**

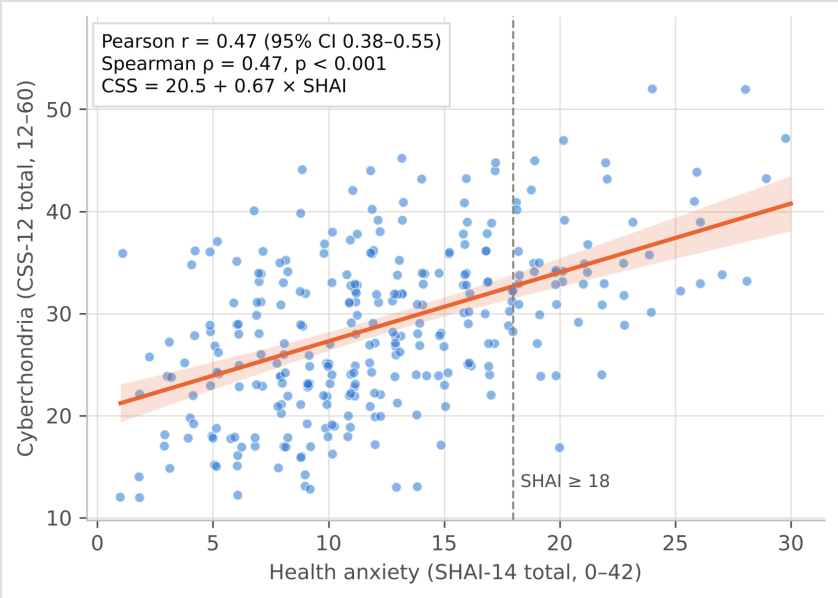
1. Two blocks: CSS-12 and SHAI-14.
2. Three rows each: configural, metric, scalar.
3. Look at ΔCFI and the Decision column: "Supported" = that step passed.

**What it shows:** CSS-12 passed all steps; SHAI-14 passed metric, not scalar; no item behaved differently.

> **Say:** *The cyberchondria questionnaire works identically in Arabic and English, so combining the two languages was justified.*

## Table 6 + Figure 5 · The main result

> **Why this was chosen:** This answers our main question (H1), so it gets both: a figure to see the pattern, a table for exact numbers. A scatter plot is the standard picture for two continuous scores.




**How to read it:**

1. Each dot = one participant (x = SHAI, y = CSS).
2. The red line = best-fit line; the shaded band = its 95% CI.
3. The dotted vertical line = SHAI 18 cut-off.
4. Table 6: r for the total and for each facet, plus adjusted β.

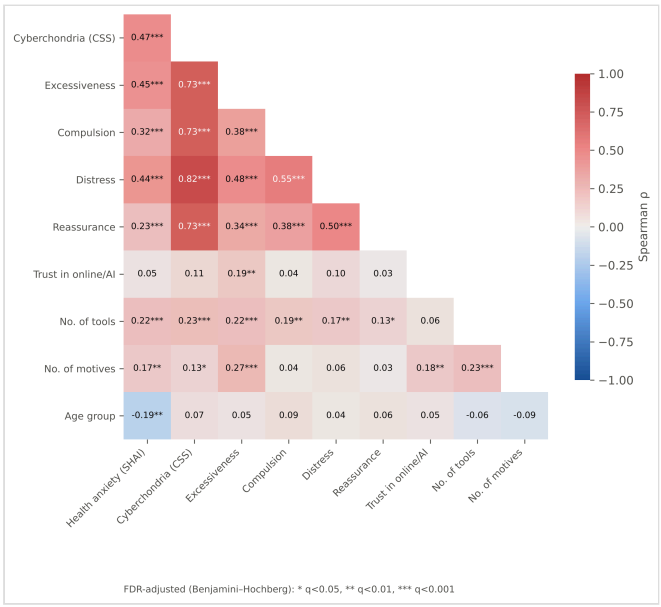
**What it shows:** Clear upward trend: r = 0.47; strongest for excessiveness (0.46) and distress (0.45).

> **Say:** *This is our main result: the more health anxiety, the more cyberchondria, r = 0.47, and it holds after adjustment.*

> **Trap:** Point at the line going up, then at the r value in the box. Do not say "causes".

## Figure 6 · Correlation heatmap

> **Why this was chosen:** We had 10 variables. Writing 45 correlations in text is unreadable; a heatmap shows them all at once, with colour for strength.




**How to read it:**

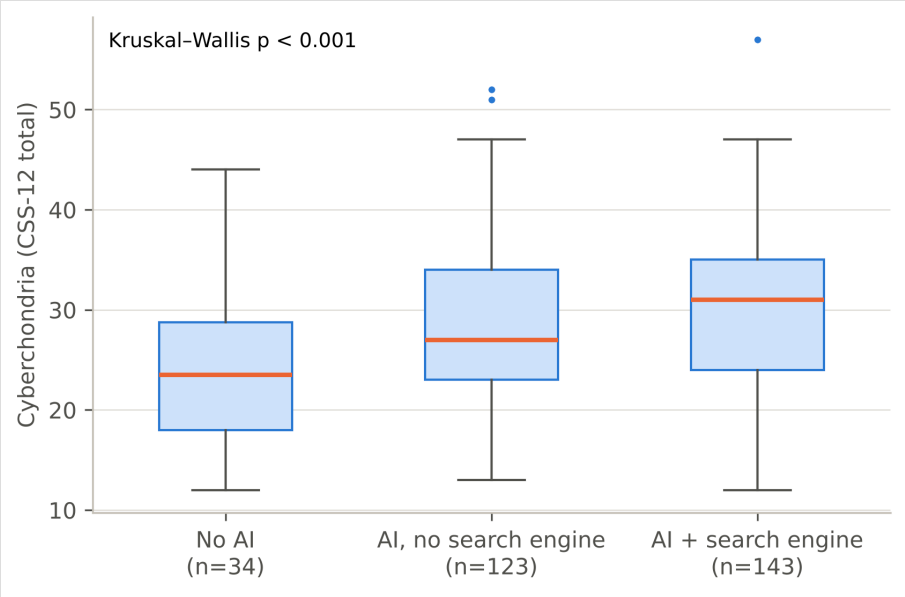
1. Each square = correlation between the row and column variable.
2. Darker red = stronger positive; blue = negative.
3. Stars = significant after FDR (* q<0.05, ** q<0.01, *** q<0.001).

**What it shows:** Health anxiety correlates with every cyberchondria facet; trust relates only to excessiveness; age is slightly negative.

> **Say:** *The heatmap shows all relationships at once: health anxiety is linked with every part of cyberchondria.*

## Table 7 + Figure 7 · Which groups score higher

> **Why this was chosen:** Table 7 compares the scores across every participant group (exact means and p/q). Figure 7 is a box plot because CSS is skewed: it shows the median and spread, not just the mean.




**How to read it:**

1. Box plot: middle line = median; box = middle half of people; whiskers = range; dots = outliers.
2. Three groups: no AI, AI only, AI + search engine.

**What it shows:** Health anxiety higher with chronic illness (14.2 vs 11.9) and in reassurance seekers (13.8 vs 10.9); CSS higher in AI users (29.5 vs 24.0), highest in AI + search engine users.

> **Say:** *People who combine AI with search engines had the highest cyberchondria, but we will see this is not robust once we adjust.*

> **Trap:** Do not over-sell the AI difference; it disappears in the strict analysis.

## Table 8 · Yes/no outcomes ($\chi^2$)

> **Why this was chosen:** Some outcomes are yes/no (elevated health anxiety, high cyberchondria). $\chi^2$ tests show which categories go with them. Only significant ones are listed to keep it short.

**How to read it:**

1. Each row: variable, % with the outcome in each group, Cramér's V (strength), p, q.

**What it shows:** With elevated health anxiety, 52.6% had high cyberchondria vs 23.0% without (OR ≈ 3.7).

> **Say:** *People with elevated health anxiety were much more likely to have high cyberchondria.*

## Table 9 · What predicts cyberchondria (regression)

> **Why this was chosen:** Group comparisons look at one factor at a time; regression tests all factors together, so we know which matter independently (H2).

**How to read it:**

1. Rows = predictors; B = CSS points per unit; β = comparable strength; p.
2. Note under the table = $R^2$ for each step.

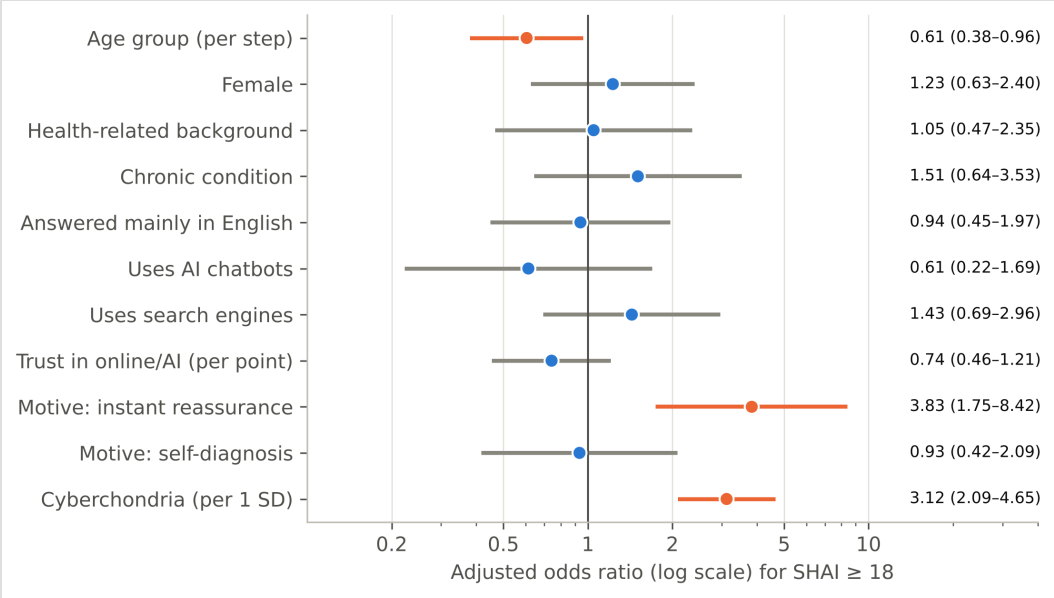
**What it shows:** Health anxiety β 0.48 (largest by far); $R^2$ rises from 0.030 to 0.317.

> **Say:** *Demographics explain only 3%; adding health anxiety explains a further 18.4%. Health anxiety is by far the strongest predictor.*

> **Trap:** AI row: significant here, but not with the strict definition (slide 23).

## Table 10 + Figure 8 · Who has elevated health anxiety

> **Why this was chosen:** For a yes/no outcome we use logistic regression; a forest plot is the standard way to show many odds ratios with CIs at once.




**How to read it:**

1. Each dot = AOR; the line = its 95% CI.
2. Vertical line at 1 = no effect. If the line crosses 1 → not significant.
3. Log scale: 0.5 and 2 are the same distance from 1.
4. Orange = significant.

**What it shows:** Cyberchondria AOR 3.12, reassurance motive 3.83, age 0.61 (protective); AUC 0.83 → 0.79.

> **Say:** *Three factors stand out: higher cyberchondria, searching for reassurance, and younger age.*

**Trap:** Only 57 events, so the reassurance odds ratio has a wide CI; say “exploratory”.

## Table 11 · Behaviour models

**Why this was chosen:** To test H4 we need adjusted models for each behaviour. A table because there are 8 outcomes × 2 scores: too many for a chart.

**How to read it:**

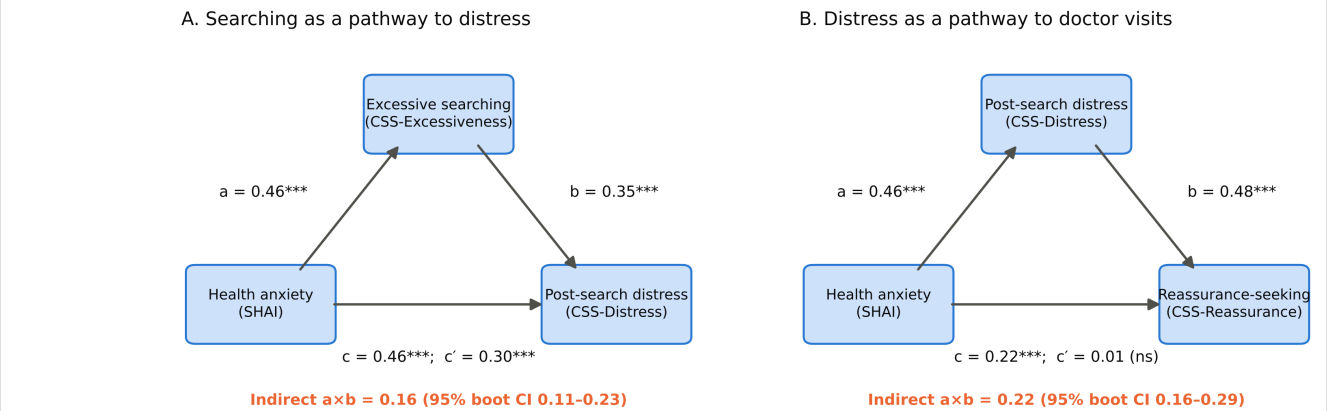
1. Each row = a behaviour; columns = AOR (95% CI) and p for SHAI and for CSS.

**What it shows:** Higher CSS → urgent visits (AOR 1.51), less ignoring (0.65), less never-disclosing (0.57, the only robust one).

**Say:** *Cyberchondria, not health anxiety, is what goes with behaviour; only the link with disclosure survived correction.*

## Table 12 + Figure 9 · Pathways (mediation)

**Why this was chosen:** To test H3 (the cycle). A path diagram is the clearest way to show a, b and direct paths; the table adds the CIs and the reverse model.




Standardised coefficients adjusted for age, gender, health background, chronic illness, trust and AI use; 5,000 bootstrap resamples. * p<0.05, ** p<0.01, *** p<0.001.

**How to read it:**

1. Boxes = variables; arrows = paths with coefficients.
2. a = X → M, b = M → Y, c′ = direct.
3. Below each diagram: the indirect effect with its CI.

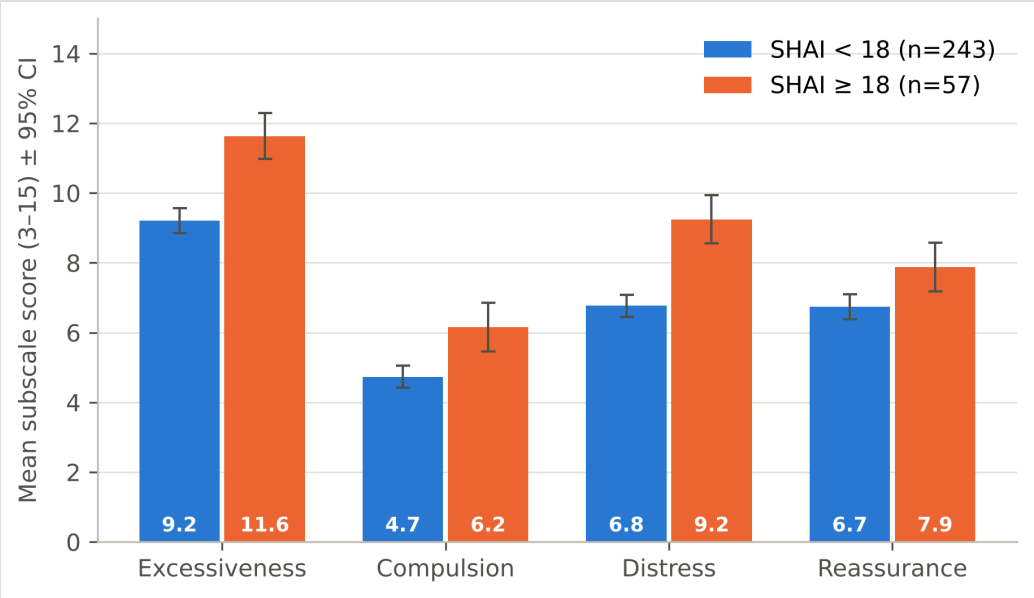
**What it shows:** Via excessive searching 0.16; via distress 0.22 (direct falls to 0.01); reverse also 0.15.

**Say:** *The pattern fits a cycle, but because the reverse direction also fits, we cannot say which comes first.*

**Trap:** Never say “proves”.

## Figure 10 · Facets by health-anxiety group

**Why this was chosen:** Shows simply that people with elevated health anxiety score higher on every facet, and which facets differ most. Bars with 95% CI are best for comparing means of two groups.




**How to read it:**

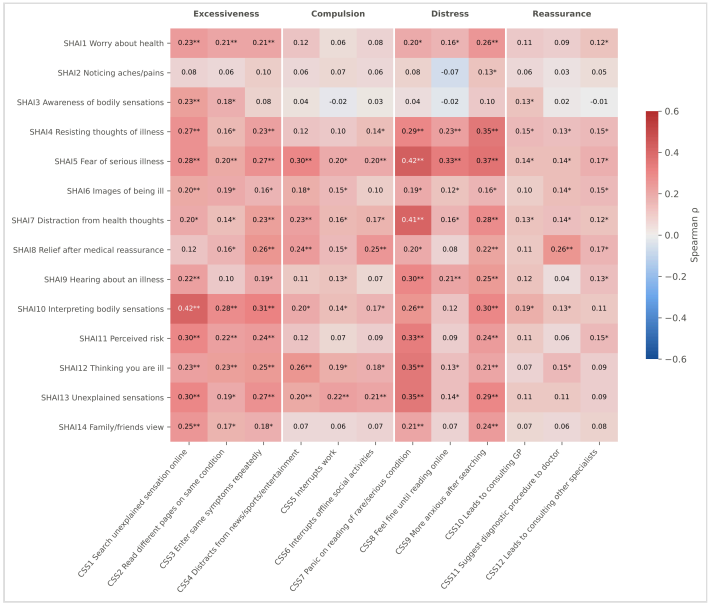
1. Blue = SHAI < 18 (243 people); orange = SHAI ≥ 18 (57 people).
2. Bar height = mean; small lines = 95% CI.

**What it shows:** Biggest gaps: excessiveness (9.2 vs 11.6) and distress (6.8 vs 9.2).

> **Say:** *People with elevated health anxiety score higher on every part of cyberchondria, most of all on excessive searching and distress.*

## Figure 11 (Appendix) · Item by item

> **Why this was chosen:** To see which exact questions drive the link (168 pairs). Only a heatmap can show 168 numbers readably, so it goes in the appendix.




**How to read it:**

1. Rows = 14 SHAI items; columns = 12 CSS items.
2. Colour = strength; stars = significant after FDR (**: also after the stricter Holm correction).

**What it shows:** 111/168 pairs significant; strongest: fear of serious illness ↔ panic on reading of a rare condition (0.43). Noticing aches is almost unrelated.

> **Say:** *What drives searching is fear and interpretation, not simply noticing symptoms.*

## Table 13 · Sensitivity analyses

> **Why this was chosen:** Committees always ask “is it robust?”. One table answers all such questions at once.

**How to read it:**

1. Each row = a check; Result; Conclusion column.

**What it shows:** r stays 0.46–0.48 whatever we exclude; only the AI effect depended on a decision.

> **Say:** *Every check left the main result unchanged.*

## Part D · Your 11 slides: what to point at and what to say

Short versions (three sentences each). The full scripts are in the speaker notes and in your Student 2 guide. Aim for 30–35 seconds per slide.

| Slide | Point at | Say (short version) |
|---|---|---|
| 13 · Results divider | Nothing to point at. | Thank you. I will present our results in six steps: who took part, how they search, the levels of health anxiety and cyberchondria, the main association, the predictors and pathways, and behaviour and AI. |
| 14 · Participants (Table 1) | The highlighted rows: gender and medical students. | Table 1 shows our 300 participants. Most were young, 62.3% aged 20 to 29, and 44.0% were medical students. In summary, a young and educated sample, which we keep in mind when reading percentages. |
| 15 · How people search (Figure 3) | The top bar of panel A (AI chatbots), then the four number boxes. | Figure 3 shows the tools people use. AI chatbots come first, 78.0%, ahead of search engines at 56.3%, and ChatGPT alone is used by 59.3%. In summary, for many people the first “doctor” is now an AI chatbot. |
| 16 · Levels & reliability (Table 3) | The first two rows, then the 19% box. | Table 3 shows the scores. 19% scored in the elevated range of health anxiety, and both questionnaires were reliable, with alpha 0.81 and 0.86. In summary, our tools measured well, in both Arabic and English. |
| 17 · Main result (Figure 5) | The rising line, then the r box. | This is our main result. Health anxiety and cyberchondria rise together, r = 0.47, and the link stays 0.48 after adjusting for everything else. In summary, hypotheses 1 and 2 are supported. |
| 18 · What predicts cyberchondria (Table 9) | The $R^2$ rows, then the 18.4% box. | Table 9 is our regression. Demographics explain only 3% of cyberchondria, but health anxiety adds another 18.4%. In summary, how anxious you are matters far more than who you are. |
| 19 · Item by item (Figure 11) | The 0.43 and 0.42 boxes, then the “≈ 0” box. | Figure 11 looks at every pair of questions. The strongest links are fear of serious illness and interpreting body sensations; simply noticing aches is almost unrelated. In summary, fear drives searching, not noticing. |
| 20 · Who has elevated health anxiety (Figure 8) | The orange dots, then the four boxes. | Figure 8 shows the odds ratios. Higher cyberchondria triples the odds, searching for reassurance nearly quadruples them, and older age is protective; the model's AUC is 0.79 after correction. In summary, the person most at risk is young and searches for reassurance. |
| 21 · Pathways (Figure 9) | Panel A indirect effect, then panel B direct effect, then the orange box. | Figure 9 shows the pathways. Part of the link runs through excessive searching (0.16), and almost all of the link to reassurance seeking runs through distress. In summary, this is compatible with a cycle, but one survey cannot prove the direction. |
| 22 · Behaviour (Figure 4) | The first bar of panel A, then the four boxes. | Figure 4 shows what people do after searching: 34.7% delay care because they feel reassured, and 31.7% never tell their doctor. Higher cyberchondria goes with more disclosure, and that result survived correction. In summary, searching changes the consultation before it begins. |
| 23 · AI chatbots | The table: inclusive vs strict rows. | Finally, AI. AI users scored 5.3 points higher, but with the strict definition the difference was only 1.5 and not significant, p = 0.31. In summary, there is no reliable independent effect of AI. I now hand over to my colleague. |

## Part E · Drills: explain it like you would to a 12-year-old

Cover the right column. Answer out loud in one or two sentences. Then check.

| Explain... | Model answer |
|---|---|
| What is a p-value? | The chance of getting a result this strong by luck if there were really no effect. Below 0.05 we call it significant. |
| What does r = 0.47 mean? | When health anxiety goes up, cyberchondria tends to go up too; a moderate link, explaining about a fifth of the differences. |
| What is a confidence interval? | The range where the true value probably lies. Ours for r is 0.38 to 0.55. |
| Why did you correct for multiple tests? | With 162 tests, about 8 would be significant just by luck; FDR correction separates real results from probable luck. |
| What is regression for? | To see which factors matter when we look at all of them together. |
| What is β? | The strength of each factor on a common scale, so we can compare them. Health anxiety had the biggest, 0.48. |
| What is an odds ratio of 3? | The odds are three times higher. Below 1 means lower odds. |
| What is AUC? | How well our model tells anxious from non-anxious people apart: 0.79 is good. |
| What is mediation? | Checking whether part of a link passes through a middle step, here searching or distress. |
| Does mediation prove a cycle? | No. The reverse also fits, and it is one survey at one time point. |
| Why did the AI effect disappear? | It depended on 32 people who named a chatbot but did not tick the box; only 34 non-users. With the strict definition it was not significant. |
| Why non-parametric tests? | Our scores were not perfectly normal; these tests do not need that. |
| What is Cronbach's alpha? | Whether the questions of a scale measure the same thing; ours were 0.81 and 0.86, good. |
| What is measurement invariance? | Checking the questionnaire measures the same thing in Arabic and English, like checking two rulers. The CSS-12 passed fully. |
| What is a sensitivity analysis? | Doing the analysis in other ways to see if the result changes. It did not. |
| Why both a table and a figure for some results? | The figure shows the pattern; the table gives the exact numbers. We did both for the most important results. |
| Why a box plot for Figure 7? | Cyberchondria scores are skewed, so the median and spread are more honest than a bar of means. |
| Why a forest plot for Figure 8? | It shows many odds ratios and their CIs at once; anything crossing 1 is not significant. |
| Why a heatmap for Figure 6? | There are 45 correlations; colour lets you see them all at once. |
| Why is Figure 11 in the appendix? | It has 168 numbers: important detail, but too much for the main chapter. |

## Cheat sheet (one page)

| What | Number | Where |
|---|---|---|
| Participants | 300; 62.3% aged 20–29; 44.0% medical students | Table 1 |
| AI chatbots / search engines | 78.0% (88.7% incl. named) / 56.3%; ChatGPT 59.3% | Figure 3 |
| Delay care when reassured / never tell the doctor | 34.7% / 31.7% | Figure 4 |
| SHAI mean / CSS mean | 12.3 / 28.9 | Table 3 |
| Reliability α | 0.81 / 0.86 | Table 3 |
| Elevated health anxiety | 19% (CI 15.0–23.8%); 32.7 / 19.0 / 7.3 / 0.7% at 15/18/22/29 | Table 4 |
| Invariance | CSS-12 scalar ✓; SHAI-14 metric ✓, scalar ×; no DIF | Table 5 |
| Main correlation | r = 0.47 (0.38–0.55); partial 0.48; $r^2$ 22% | Table 6, Figure 5 |
| Facets | excess 0.46, distress 0.45, compulsion 0.25, reassurance 0.24 | Table 6 |
| Regression $R^2$ | 0.030 → 0.133 → 0.317; $\Delta R^2$ 18.4%; β 0.48; B 0.68 | Table 9 |
| Logistic | AOR 3.12 / 3.83 / 0.61; AUC 0.83 → 0.79 | Table 10, Figure 8 |
| Behaviour (robust) | CSS → never disclose AOR 0.57, q 0.012 | Table 11 |
| Mediation | via excess 0.16 (0.11–0.23); via distress 0.22, direct 0.01; reverse 0.15 | Table 12, Figure 9 |
| AI effect | +5.3 inclusive vs +1.5 strict (p 0.31); 32 named-not-ticked; 34 non-users | Table 9, slide 23 |
| Item level | 111/168 FDR; top ρ 0.43 | Figure 11 |
| Robustness | r 0.46–0.48 in every check | Table 13 |

> **Say:** *Health anxiety and cyberchondria travel together, r = 0.47; AI chatbots are the main source but have no reliable independent effect; and searching changes the consultation before it begins.*